

Cortical Decoders Trained on Other Individuals Age Slower Than Your Own

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Abstract

With adapting brain–computer interface (BCI) technology, neural decoding algorithms are one of the most essential parts of being able to understand people’s brains. Decoders must work for months, and recalibrating them per subject has high subject costs. A decoder fit to a singular animal’s cortex drifts, losing accuracy over days, and usually the remedy is recalibrating on fresh data from that same subject. We report a different kind of robustness: a decoder built from one other individual, or pooled from a set of other individuals, ages more slowly than the subject’s own. On the open BraiDyn-BC cohort (25 mice, 44 Allen-atlas cortical parcels, a ~ 15 -day operant protocol) we decode four task events and measure, per mouse, the AUC lost over two weeks by its own early-week decoder against one pooled over the other 24 mice and one built from a single other mouse. The personal decoder ages more by $+0.017$ AUC ($p = 5 \times 10^{-5}$, 73 of 100 mouse–event pairs). This also is not a data-volume effect: at a matched training-event count a decoder from a single other mouse still ages less. Hence we conclude this is rather an effect of not overfitting an individual’s drift-prone idiosyncrasies. It also survives a 1-D CNN and a GRU. The effect holds past this cohort and recording modality. On an independent two-photon dataset from a different laboratory – Allen Visual Behavior, 23 mice, single-plane VISp populations decoding a stimulus change – the same session-held-out protocol gives $+0.023$ AUC ($p = 0.003$, 74% of mice), and the count-matched control reproduces there as well (self – one other mouse = $+0.011$ AUC). The early-week accuracy cost of decoding across individuals is repaid as late-week stability, which suggests population priors as an alternative to per-subject recalibration. A second offering is the session-held-out protocol under which all of this is measured: because only the within-subject arm can share a recording session with its test data, a trial-level split inflates that arm alone, and holding out whole sessions is what allows this form of robustness to be reported reliably. Code and a streamed-feature pipeline are released.

1. Introduction

Neural decoding is one of the most essential ways to understand neural interfaces and create BCI technology, and a decoder that is deployed is a decoder that has to keep working: over months, on the same individual, without fresh labelled data arriving to prop it up. Representational drift stands directly against that. The reorganization of an animal’s neural code over days with fixed behavior is now documented across sensory, motor, and association cortex and in the hippocampus (Rule et al., 2020; Driscoll et al., 2017; Schoonover et al., 2021). This has a stark impact on neural decoding: a decoder trained on an animal today degrades on that same animal weeks later. The dominant response is recalibration, periodically re-fitting the decoder on fresh within-subject data (Degenhart et al., 2020; Jarosiewicz et al., 2015), which requires ongoing labelled data from the individual and treats each subject in isolation.

The code’s cross-individual structure offers an alternate route to temporal robustness, and the field already relies on it. Cortex-wide behavioral representations are considerably shared among individuals (Musall et al., 2019; MacDowell & Buschman, 2020; MacDowell et al., 2024); latent dynamics are preserved across animals well enough that a decoder fit to one transfers to another (Safaie et al., 2023); and pooled decoders already read out behavior from unseen individuals (Hosseini et al., 2026; Zhang et al., 2025; Peterson et al., 2021). Methods such as Meta-AlignNN (Zou et al., 2025) are built on that premise. While people currently know that conservation across individuals is established, and that pooling works at a fixed time, every one of those results is measured at a single moment. We offer pooling as a solution to a different problem: the decay of decoding accuracy over time.

Our core prediction is that a decoder that never saw the subject cannot fit to that subject’s drift-prone idiosyncrasies, so it should age more slowly, and the way we measure that difference is as an asymmetry between the two arms. This prediction, to our knowledge, has not been examined.

We test it on BraiDyn-BC (Kondo et al., 2025), a widefield calcium-imaging resource whose ~ 15 -day operant protocol and shared 44-region Allen parcellation make within-subject and cross-subject decoders directly comparable across time. Our contributions:

- **Pooling resists drift.** Across a two-week period the population-pooled decoder loses less accuracy than the individual’s own, with a paired asymmetry of $+0.017$ AUC ($p = 5 \times 10^{-5}$, 73 of 100 pairs). The size of this effect varies among the four events, but those are chained within a trial and have highly differing counts, so we treat the pooled estimate as the result and leave the per-event ranking uninterpreted.
- **It is not a data-volume artifact.** At a matched training-event count a decoder trained on one other mouse ages less than one trained on the subject’s own, by $+0.010$ AUC on average. The gain therefore comes from training on individuals other than the subject, not from the larger training set pooling usually brings, and training on many other mice rather than one lowers aging further on all four events.
- **It replicates on an independent dataset.** On the Allen Visual Behavior two-photon cohort the same protocol gives $+0.023$ AUC ($p = 0.003$, 74% of 23 mice), with the count-matched ordering also reproduced.
- **Session-level validation is load-bearing.** As trials within a recording session are dependent, a trial-level split inflates the personal decoder but not the pooled one, one-sidedly inflating the asymmetry. We quantify that inflation, and on lever pull it accounts for the entire apparent effect. For longitudinal decoding comparisons we recommend session-held-out validation as a default.

2. Data and methods

BraiDyn-BC (Kondo et al., 2025) (DANDI:001425) provides dorsal-cortex $\Delta F/F$ parcellated into 44 Allen Common Coordinate Framework regions for 25 mice, each completing ~ 15 daily operant sessions over ~ 3 weeks. For each onset of four events (lever pull, lick, reward, tone) we build a 44-dimensional evoked feature and decode it against matched negatives with ℓ_2 -regularized logistic regression, scored by area under the receiver-operating

characteristic curve (AUC). Sessions are grouped into an early block (days 1–5) and a late block (days 11–15). The personal decoder leaves out one whole early *session* at a time; the pooled decoder trains on the other mice’s early blocks. Aging is the accuracy lost as testing moves from early to late with training fixed, and the per-mouse asymmetry is personal aging minus pooled aging, positive when the personal decoder ages more. Significance is a paired sign-flip permutation test across mice (20,000 permutations), with cluster-bootstrap confidence intervals. Holding out whole sessions rather than trials is essential rather than incidental: only the within-subject arm can share a session between train and test, so a trial-level split inflates that arm alone. The external validation cohort is Allen Visual Behavior two-photon (Allen Institute for Brain Science, 2023), for which the absence of any neuron correspondence across animals forces a permutation-invariant 27-dimensional summary of the population response. Appendix C gives both in full.

3. Results

For each mouse we keep the test data fixed (first the mouse’s held-out early sessions, then its late block) and change only the source of the training data: either the mouse itself or the other 24 mice. Across all 100 mouse–event pairs the asymmetry is +0.017 AUC ($p = 5 \times 10^{-5}$), with the personal decoder aging more in 73% of pairs; the per-event breakdown is in Table 2 of Appendix B.

The effect differs among the four events: it is largest for reward (+0.029, $p = 0.0005$) and lick (+0.020, $p = 0.002$), marginal for tone (+0.015, $p = 0.08$), and absent for lever pull (+0.003, $p = 0.37$). While we report this variation, we do not interpret it, as the four events all fall within roughly the same two seconds of a trial. Accordingly, the pooled estimate over all 100 pairs is the claim we defend, and the per-event column should be read as data rather than as four separate findings.

A pooled decoder contains $\sim 24\times$ more training events than a personal decoder, so its stability could reflect data volume rather than pooling. We test this with a count-matched control. For each held-out early session s of each mouse we set the training-set size to n , the number of events in that mouse’s remaining early sessions, and compare three decoders: *self*, trained on those n events; *one*, trained on n events from a single other mouse; and *many*, trained on n events drawn across the other 24 mice. All three are scored on the same held-out session s and the same late block, so they differ only in whose data they were fit on.

For the events where there was an asymmetry, the count-matched ordering reproduces it. The self decoder ages more than a decoder trained on one other mouse by +0.010 AUC on average, and by +0.007 for lick, +0.017 for reward and +0.006 for tone individually. As both decoders see the same number of training events, this is not a data-volume artifact.

To further test whether the asymmetry is a property of that one cohort, modality and task, we repeat the analysis on the Allen Visual Behavior two-photon dataset (Allen Institute for Brain Science, 2023): a different laboratory, cellular two-photon rather than mesoscale widefield imaging, a visual change-detection task rather than a cued lever pull, and a 27-dimensional permutation-invariant population summary rather than a 44-parcel atlas feature. The asymmetry reproduces at a comparable magnitude (Table 1): across 23 mice the personal decoder ages by +0.024 AUC and the pooled decoder by +0.001, an

Table 1: The aging asymmetry on the original cohort and on the external validation dataset. Aging is AUC lost from early to late test with training held fixed; asymmetry is personal – pooled aging, positive when the subject’s own decoder ages more. Both rows use the same session-held-out protocol and the same count-matched control. *p*: paired sign-flip permutation. Per-arm aging for BraiDyn-BC is broken out by event in Table 2.

	Personal	Pooled	Asymmetry (95% CI)	<i>p</i>	own ages more
BraiDyn-BC, widefield, 25 mice	–	–	+0.017 [0.010, 0.024]	5×10^{-5}	73%
Allen VB, two-photon, 23 mice	+0.024	+0.001	+0.023 [0.010, 0.037]	0.003	74%

Count-matched control, aging at equal training-event count. BraiDyn-BC: self – one other mouse = +0.010; many others age less than one on all four events. Allen VB: self +0.024, one +0.013, many +0.003; self – one = +0.011, many – one = –0.010.

asymmetry of +0.023 [0.010, 0.037], $p = 0.003$, with the mouse’s own decoder ageing more in 74% of animals. The count-matched control reproduces as well, at +0.011 [0.001, 0.022] for self minus one other mouse, though on 23 animals it is marginal by the sign-flip test ($p = 0.054$). The mechanism is not identical in the two cohorts – on BraiDyn-BC the gap narrows because the pooled decoder gains, here because the personal decoder loses – and the shared finding is that the personal decoder ages faster. The result does not depend on where the session-inclusion threshold is drawn, giving +0.022 to +0.024 across thresholds of three to six sessions per mouse. On this cohort the trial-level split *lowers* the asymmetry to +0.014 ($p = 0.07$): trial shuffling still inflates the personal decoder by +0.015, but scoring one pooled AUC across sessions rather than averaging per-session values costs –0.030 ($p < 10^{-4}$) and dominates.

4. Discussion

A subject’s own decoder overfits idiosyncratic features of its early sessions, some of which later drift. A decoder that has never seen the subject must instead rely on components of the code that are shared across individuals and more stable over time. The count-matched control supports this reading directly: at equal data, other-individual data ages less than the subject’s own. Unlike within-subject recalibration, this requires no newly labelled data from the individual at test time – the practical point, since obtaining that data is the expensive part of deploying a decoder over months.

The validation protocol is worth stating explicitly: a trial-level split inflates the personal decoder’s within-block accuracy by up to +0.013 AUC and leaves the pooled decoder untouched, and on lever pull that inflation accounts for the entire apparent effect (+0.014, $p = 0.001$, becoming +0.003, $p = 0.37$). The practical consequences are collected as a checklist in Appendix A.

Code and data availability. Both cohorts are public (DANDI:001425; Allen Brain Observatory). The analysis code and streaming feature pipeline are released; the repository link is withheld here to preserve anonymity and will be given in the camera-ready version.

Limitations. The four events are not four independent behaviors but four thresholded channels of one cued lever-pull trial, so we cannot always say which behavior the effect belongs to. We measure at the mesoscale, and the two-photon replication reaches single neurons only at one depth in one visual area. The trial-level inflation above is a caution for comparisons of this shape rather than a limitation of this work: every number we report is measured after holding out whole sessions.

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Appendix A. Best-practice checklist

Best practice for longitudinal within- versus cross-subject comparisons

1. **Hold out whole recording sessions, not trials.** Only the within-subject arm can share a session between train and test. Here a trial-level split inflated it by up to +0.013 AUC and on lever pull accounted for the entire apparent effect (+0.014, $p = 0.001 \rightarrow +0.003$, $p = 0.37$).
2. **Count-match the training sets, and never score an arm on the block it was fit on.** A pooled decoder holds $\sim 24\times$ more events; scoring *self* on its own training block is not a cross-validated quantity and reverses the ordering the control exists to establish.
3. **Average per-session scores rather than pooling one score across sessions,** and report each arm’s aging separately. On the Allen cohort a single pooled AUC cost -0.030 ($p < 10^{-4}$) and dominated the leakage term it was confounded with.

Appendix B. Per-event results

Table 2: Personal vs. pooled aging across days (AUC lost from early to late test, holding training fixed), per event, $n = 25$ mice, with whole early sessions held out. Asymmetry = personal – pooled aging; a positive value means the mouse’s own decoder ages more. p : paired sign-flip permutation. The rightmost column gives the same asymmetry under a trial-level split, the protocol this analysis replaces.

Event	Personal	Pooled	Asymmetry (95% CI)	p	own ages more	trial-split
Lever-pull	−0.022	−0.025	+0.003 [−0.004, 0.011]	0.37	64%	+0.014
Lick	+0.012	−0.009	+0.020 [0.008, 0.032]	0.0023	80%	+0.028
Reward	+0.017	−0.012	+0.029 [0.014, 0.046]	0.0005	80%	+0.032
Tone	+0.016	+0.001	+0.015 [0.001, 0.032]	0.077	68%	+0.022
Pooled			+0.017 [0.010, 0.024]	5×10^{-5}	73%	+0.024

Appendix C. Data and methods in full

C.1. Datasets and preprocessing

BraiDyn-BC (Kondo et al., 2025) (DANDI:001425, CC-BY) provides, for each mouse and session, hemodynamically corrected dorsal-cortex $\Delta F/F$ traces parcellated into 44 Allen Common Coordinate Framework regions at ~ 30 Hz, together with time-aligned behavioral channels. The dataset includes 25 mice, each of which completed ~ 15 daily operant sessions over ~ 3 weeks. We stream the 44-region traces and behavioral channels without downloading the raw movies.

The two cohorts used in this paper were built for different purposes, and the ways they differ are what make the comparison informative rather than redundant. BraiDyn-BC is densely longitudinal, with roughly fifteen sessions per mouse over three weeks, and it supplies a shared coordinate system: the 44 Allen parcels mean that region k in one mouse is the same region in every other, so a pooled decoder can be fit directly in a common feature space. The Allen Visual Behavior cohort has neither property. Its longitudinal subset is sparser, and because it is cellular two-photon imaging there is no correspondence between animals at all – one mouse’s cell 17 is unrelated to another’s – so a common feature space has to be constructed rather than assumed (Section C.4). The recording modality, the task, and the decoded contrast all differ as well. A result that survives both is therefore not a property of one atlas, one instrument, or one behavior.

C.2. Features and decoders

We detect the onset of four events (lever pull, lick, reward, and tone) as rising edges, with a 1 s refractory period. For each onset we construct a 44-dimensional evoked feature by subtracting the mean pre-onset baseline (-0.5 – 0 s) from the mean post-onset $\Delta F/F$ response (0 – 1 s) in each parcel. Negative samples are drawn from matched-count windows more than 2 s from any onset *of the same event*; they are not screened against the other three events. Every classification is a binary decode of one event against these matched negatives, scored by AUC. The four decoders are therefore four independent binary detectors, not a four-way classifier: nothing in this paper asks a model to distinguish one event from another.

The primary decoder is a standardized ℓ_2 -regularized logistic regression on the 44-dimensional evoked feature. This is a linear readout of the parcel-space code: it recovers the component of behavior that is linearly present in the population average, and it is the natural comparison to prior cross-subject decoders, which are overwhelmingly linear (Melbaume et al., 2022; Peterson et al., 2021). Every condition below uses this same readout, so the only variable is which data the decoder was trained on.¹

1. The result is not a property of the linear readout. Repeating the analysis with a 1-D convolutional network over time and a GRU over the frame sequence (full evoked window, -0.5 to $+1.0$ s, 45×44), under the same session-held-out protocol, the asymmetry survives for both events that carry it: lick gives $+0.020$, $+0.012$ and $+0.010$ and reward $+0.029$, $+0.015$ and $+0.012$ under the linear, CNN and GRU readouts, all significant. Under a trial-level split the nonlinear estimates appear *larger* instead, which is the trial-level inflation: a higher-capacity model exploits it more effectively than a linear one.

C.3. Temporal blocks, aging, and the session-held-out protocol

We group each mouse’s sessions into an early block (days 1–5) and a late block (days 11–15) and report AUC under four conditions. The personal decoder is estimated by leaving out one whole early *session* at a time: for each early session s we train on the mouse’s remaining early sessions and score the resulting decoder both on s (within-mouse same-block, WS) and on the late block (within-mouse across-block, WX), then average over s . Because WS and WX come from the same fitted decoders, they differ only in which data they are tested on. The pooled decoder is trained once on the other mice’s early blocks and scored on the same held-out sessions s (LS) and on the late block (LX).

Holding out sessions rather than trials is essential and not a detail. Trials within one recording share illumination and hemodynamic state, slow arousal drift, and temporal autocorrelation, so a trial-level split lets the personal decoder train and test inside the same session. The pooled decoder is trained on other animals entirely and cannot benefit this way, so the inflation is one-sided and lands directly on the quantity of interest; Table 2 reports what it costs per event.

Aging is the accuracy lost when testing moves from the held-out mouse’s early data to its late block while training stays fixed: personal aging is $WS - WX$ and pooled aging is $LS - LX$. The per-mouse aging asymmetry is $(WS - WX) - (LS - LX)$; a positive value means the personal decoder ages more. We assess significance with a paired sign-flip permutation test across mice (20,000 permutations) and compute 95% confidence intervals with a cluster bootstrap over mice.

C.4. The external validation cohort

The external validation repeats the analysis on the Allen Visual Behavior two-photon dataset (Allen Institute for Brain Science, 2023). We select containers spanning at least 14 days whose experiments are all in VISp within a single 50 μm depth bin, which is the cross-subject correspondence available here; the largest such band, 150–199 μm , holds 25 mice and 184 experiments. We require at least 15 segmented cells and 40 usable trials per experiment, leaving 166 experiments and 23 mice with at least four sessions. Sessions are streamed from S3 rather than downloaded.

As different animals have no neuron correspondence and the longitudinal subset is single-plane, averaging over the field of view would give a one-dimensional feature. The representation that is comparable across animals is therefore a permutation-invariant summary of the population response: per neuron we take the mean $\Delta F/F$ over 0–1 s minus the mean over -0.5 –0 s, then describe the distribution of those evoked responses by 25 quantiles plus its mean and standard deviation, giving a fixed 27-dimensional feature regardless of how many cells were segmented. The decoded contrast is `is_change`, a real image change against a sham catch trial, which is a stimulus contrast rather than a reward- or lick-coupled one. Each mouse’s sessions are split by date into early and late halves, and the personal decoder, the pooled decoder, aging, the asymmetry and the count-matched control are all defined exactly as above.